Robin Lange

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SUMMARY

International Mechanical Engineering Honors student from Germany, currently studying at Georgia Southern University in the United States on an F1 Student Visa. Research experience in collaborative autonomy, perception, and predictive control across simulation and hardware. Interested in combining learned models with planning and control for reliable autonomy on physical robots.

EDUCATION

B.S. Mechanical Engineering, Vehicle Design and Testing Emphasis

GPA: 4.00/4.00

Georgia Southern University **Aug 2023 – May 2027 · Statesboro, GA**

Honors College Scholar, President's List, Allen E. Paulson Scholar, Tau Beta Pi

Honors Thesis (in progress): Reliability-Gated Lead-Vehicle Prediction for Safety-Supervised Adaptive Cruise Control

Relevant Coursework: Mechatronics, Mechanism Design, Machine Design, Dynamics, Mechanics of Materials, Measurement & Instrumentation Lab, Vehicle Design and Testing, Numerical Methods, Differential Equations, Calculus I–III, Program Design, Circuit Analysis, Logic Circuit Design, FEA, and Engineering Research.

EXPERIENCE

Undergraduate Student Researcher

Intelligent Vehicles Navigation Research Lab, Georgia Southern University **Oct 2023–Present · Statesboro, GA**

- Investigating when safety-supervised MPC should trust learned lead-vehicle forecasts; trained a GRU with 27.5% lower held-out forecast RMSE than physics-based extrapolation and evaluated reliability gating and switching-induced control transients.
- Built an Isaac ROS cuVSLAM pipeline on Jetson/RealSense UAV hardware; measured 14.5 cm localization RMSE in a 110-second ground trial over valid-reference intervals.
- Contributed LSTM actuator modeling and ONNX/Gazebo integration to an SAE study reporting 36.3% lower wheel-velocity RMSE; designed parameterized car/truck AEB scenarios in OpenSCENARIO/AVL.

Collaborative Robotics Autonomy Intern

IHMC National Center for Collaborative Autonomy **May – Aug 2026 · Pensacola, FL**

- Developed a distributed Rust/Python autonomy platform for heterogeneous robot teams, integrating Joint Activity Graphs, capability-based task allocation, Zenoh communications, command validation, and runtime supervision.
- Designed hardware-agnostic interfaces and robot adapters; implemented staged Spot grasping with motion-capture positioning and arm-camera refinement, resolving sensing-to-robot coordinate-frame inconsistencies.
- Built Bevy/Avian simulation and hybrid digital twins for simulation-to-hardware validation; demonstrated coordinated collection-and-delivery missions with two autonomous Boston Dynamics Spots and a human-operated Clearpath Husky across more than a dozen hardware trials.

Vehicle Testing Research Intern

CARISSMA Institute of Safety in Future Mobility **Jul – Aug 2025 · Ingolstadt, Germany**

- Designed and built a truck-rear soft target for repeatable vehicle safety pre-crash testing, linking mechanical design with camera, LiDAR, and radar detectability.
- Analyzed measured and MAN TruckScenes reference radar/LiDAR data; investigated dominant trailer-underbody multipath returns in the examined cases and limits of reproducing them with a compact target.
- Fine-tuned YOLO26l on BDD100K-derived images for truck-rear detection (mAP@50: 0.982); used the Vehicle Orientation Dataset and custom imagery to evaluate orientation detection and a two-stage truck-rear/underride-guard pipeline with visual-prompt YOLOe.

Autonomous Systems and AI Intern

Fraunhofer Institute for Transportation and Infrastructure Systems **Jun – Aug 2024 · Ingolstadt, Germany**

- Developed hybrid state-machine and behavior-tree mission AI for ALBACOPTER, combining explicit transition guards with reactive actions for extensible mission sequencing and emergency interruption.
- Implemented the mission logic using Python py_trees and ROS 2 Humble nodes; validated all defined state transitions and trigger-driven emergency preemption in Fraunhofer's in-house flight simulator.
- Benchmarked state-machine, behavior-tree, and hybrid designs by execution time and code size; selected the hybrid for extensibility despite the pure behavior tree's faster execution.

PUBLICATIONS

Design and Validation of a Soft Crash Target for Multimodal Sensor-Based Truck Underride Testing
Lange, R., Sezgin, F., Inderst, M., Sutton, T., Lugner, R., Brandmeier, T., and Soloiu, V. *Manuscript in preparation.*

Data-Driven Calibration of Vehicle Simulations for Accurate Control Response in Development and Testing
Soloiu, V., Sutton, T., Mehrzed, S., **Lange, R.**, Zimmerman, C., and Peralta Lopez, G. *SAE Technical Paper 2026-01-0058 (2026).* doi:10.4271/2026-01-0058

Warehouse Platooning Framework Using a Sensor-Rich Leader and Minimal Perception Follower
Soloiu, V., Sutton, T., **Lange, R.**, and Zimmerman, C. *SSRN Preprint (2026; not peer reviewed).* doi:10.2139/ssrn.7067358

Development of a Rule-Based Lidar Lane-Keeping Assistance System for Autonomous Vehicles Applications
Soloiu, V., Mehrzed, S., Kroeger, L., Pierce, K., Sutton, T., and **Lange, R.** *SAE Technical Paper 2025-01-8025 (2025).* doi:10.4271/2025-01-8025

RESEARCH PRESENTATIONS & POSTERS

Intelligent UAV Navigation with V-SLAM and Behavior Trees for Industry 5.0 Applications
Southern Regional Honors Council Conference, Montgomery, AL (2026); related UAV inspection poster at Georgia Southern Research Symposium.

3D-Space Hand Trajectory Tracking for Intuitive UAV Control
Southern Regional Honors Council Conference, Chattanooga, TN (2025); related hand-guided drone poster at Georgia Southern Research Symposium.

Analyzing the Performance of 3D LiDAR and Camera-Based Object Detection Systems and Their Fusion Using AI for Autonomous Systems
Coauthored research poster, Georgia Southern Research Symposium (2024).

Determining Aircraft Final Approach Altitude and Attitude Using a Machine Learning-Based Sensor Fusion System
Coauthored research poster, Georgia Southern Research Symposium (2024).

AWARDS & HONORS

Gulfstream Innovation, Engineering and Flight Excellence Awards — Third Place, Undergraduate
Georgia Southern University, Allen E. Paulson College of Engineering & Computing Research Symposium · 2026

Honorable Mention, Engineering Division Student Research Video Competition
Council on Undergraduate Research (CUR), Engineering Division · 2026

Georgia Power Innovation Undergraduate Awards — Second Place
Georgia Southern University, Allen E. Paulson College of Engineering & Computing Research Symposium · 2025

INVOLVEMENT

Honors Peer Mentor · Georgia Southern Honors College · August 2026 – Present
Physics Teaching Assistant and Tutor · Georgia Southern Physics Department · January 2025 – Present
Community Leadership and Service · Southern Leaders · August 2023 – Present
First-Year Experience Peer Leader · Georgia Southern First & Second Year Experience · August – December 2024
Programming for Engineers Lab Assistant · Georgia Southern Electrical Engineering · January – May 2024

SKILLS

Robotics & Autonomy: ROS 2, Nav2, MAVROS, ArduPilot, Zenoh, Joint Activity Graphs, behavior trees/state machines, multi-robot task allocation, coordinate transforms, inverse kinematics.

Hardware & Engineering: Boston Dynamics Spot, Clearpath Husky, TurtleBot 4 Lite, RealSense cameras, Ouster 3D LiDAR, 4D radar, OptiTrack/Motive, motor/encoder integration, LabVIEW, SolidWorks, ANSYS (FEA/CFD).

Perception & Learning: Isaac ROS cuVSLAM, visual SLAM, LiDAR/RGB-D perception, point-cloud processing, sensor fusion, custom YOLO training/evaluation, GRU/LSTM modeling.

Control & Simulation: PID control, safety-supervised MPC, reliability-gated preview, MATLAB/Simulink, AVL Scenario Simulator, Gazebo, Bevy/Avian3D, CARLA replay, MCAP/Foxglove.

Programming & Development: Rust, Python, MATLAB, C++, Arduino, Git, Linux, CUDA, PyTorch, Docker, CI/CD (GitHub Actions/GitLab), NVIDIA Jetson Orin AGX /Nano.

Languages: German (native), English (fluent), French (intermediate).